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from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split, GridSearchCV

from sklearn.linear_model import LinearRegression, Ridge, Lasso

from sklearn.metrics import mean_squared_error, r2_score

from sklearn.pipeline import Pipeline

from sklearn.preprocessing import StandardScaler


# Load the Diabetes dataset

data = load_diabetes()

X = data.data

y = data.target


# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# Standard Linear Regression

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)


# Make predictions

linear_prediction = linear_model.predict(X_test)


# Calculate performance metrics

linear_mse = mean_squared_error(y_test, linear_prediction)

linear_r2 = r2_score(y_test, linear_prediction)


# Ridge Regression with cross-validation

ridge_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("ridge", Ridge())
])


# Define alpha values for tuning

ridge_params = {
    "ridge__alpha": [0.01, 0.1, 1, 10, 100]
}


# Perform 5-fold cross-validation
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ridge_grid = GridSearchCV(
    ridge_pipeline,
    ridge_params,
    cv=5,
    scoring="neg_mean_squared_error"
)

# Train Ridge model
ridge_grid.fit(X_train, y_train)

# Make predictions
ridge_prediction = ridge_grid.predict(X_test)

# Calculate performance metrics
ridge_mse = mean_squared_error(y_test, ridge_prediction)
ridge_r2 = r2_score(y_test, ridge_prediction)

# Lasso Regression with cross-validation
lasso_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("lasso", Lasso(max_iter=10000))
])

# Define alpha values for tuning
lasso_params = {
    "lasso__alpha": [0.01, 0.1, 1, 10, 100]
}

# Perform 5-fold cross-validation
lasso_grid = GridSearchCV(
    lasso_pipeline,
    lasso_params,
    cv=5,
    scoring="neg_mean_squared_error" )

# Train Lasso model
lasso_grid.fit(X_train, y_train)

# Make predictions
lasso_prediction = lasso_grid.predict(X_test)

# Calculate performance metrics

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lasso_mse = mean_squared_error(y_test, lasso_prediction)

lasso_r2 = r2_score(y_test, lasso_prediction)


# Display results

print("STANDARD LINEAR REGRESSION")

print("MSE:", linear_mse)

print("R2:", linear_r2)

print("\nRIDGE REGRESSION")

print("Best Alpha:", ridge_grid.best_params_["ridge__alpha"])

print("MSE:", ridge_mse)

print("R2:", ridge_r2)

print("\nLASSO REGRESSION")

print("Best Alpha:", lasso_grid.best_params_["lasso__alpha"])

print("MSE:", lasso_mse)

print("R2:", lasso_r2)
```

STANDARD LINEAR REGRESSION

MSE: 2900.193628493482

R2: 0.4526027629719195

RIDGE REGRESSION

Best Alpha: 10

MSE: 2875.7787184218428

R2: 0.4572109567780849

LASSO REGRESSION

Best Alpha: 1

MSE: 2824.568094049959

R2: 0.46687670944102466